

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 November 2020

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2020 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [].

INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

1. (a) Name two structures found in a plant cell but not in an animal cell. [2]<<
>>(b) State the function of the cell membrane. [2]<<
>>(c) Describe how to prepare a slide of onion epidermis to view under a light microscope. [3]<<
>>(d) Explain why muscle cells contain many mitochondria. [3] **[10]**

2. (a) Write the balanced symbol equation for photosynthesis. [3]<<
>>(b) Name the green pigment and the organelle where it is found. [2]<<
>>(c) State two factors that can limit the rate of photosynthesis. [2]<<
>>(d) Describe a test for starch in a destarched leaf that has been exposed to light. [3] **[10]**

3. (a) Name three components of blood. [3]<<
>>(b) State the function of red blood cells. [2]<<
>>(c) Describe how a white blood cell can destroy a bacterium. [3]<<
>>(d) Explain why plasma is needed. [2] **[10]**

4. (a) Name the type of pathogen that causes malaria. [1]<<
>>(b) Name the vector of malaria. [1]<<
>>(c) Describe how malaria is transmitted from an infected person to a healthy person. [4]<<
>>(d) Suggest two methods of controlling the spread of malaria. [4] **[10]**

5. (a) Describe how an ionic bond forms between sodium and chlorine. [3] <<
>> (b) Describe a covalent bond. [2] <<
>> (c) Explain why sodium chloride has a high melting point. [3] <<
>> (d) Explain why chlorine gas, Cl_2 , has a low melting point. [2]
[10]
6. (a) State the pH of a strong acid and of a neutral solution. [2] <<
>> (b) Write a word equation for an acid reacting with a metal. [2] <<
>> (c) Write a word equation for an acid reacting with an alkali. [2] <<
>> (d) Describe a laboratory test for hydrogen gas. [2] <<
>> (e) Name the salt formed from hydrochloric acid and sodium hydroxide. [2]
[10]
7. (a) State the principle of conservation of energy. [2] <<
>> (b) A 2 kg mass is lifted through 10 m. Take $g = 10 \text{ N/kg}$. Calculate the gain in gravitational potential energy. [3] <<
>> (c) Name the energy transfer that occurs in a battery-powered torch when it is switched on. [2] <<
>> (d) Explain how heat is transferred through the metal wall of a kettle by conduction. [3] **[10]**
8. (a) State the rule for attraction and repulsion of magnetic poles. [2] <<
>> (b) Name two magnetic materials. [2] <<
>> (c) Describe how to plot the magnetic field of a bar magnet using a plotting compass. [3] <<
>> (d) Explain how an electromagnet can be made stronger. [3] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).